UNIT CIRCLE VOLVELLE & TRIGONOMETRIC CALCULATOR

Special Right Triangles, Radians, Coordinates & Wave Projections

CARD 08 / 10 • VOLV-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · 6·Algebra · 7·Probability · **8·Volvelle** · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

CIRCULAR COORDINATES & DYNAMIC PROJECTIONS

Trigonometry is circular motion, not triangles.
On unit circle (r = 1), coordinates are trig values:
(X, Y) = (cos θ, sin θ)
Rotating disc projects angle θ onto cos/sin!

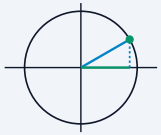
CALIBRATED 84.00 MM RADIUS DATUM

- Outer Ring: 84.00 mm ⌀ (Matches Fraction 1 Whole)
- Dual Scales: Degrees (0°–360°) and Radians (0–2π)
- Special Angles: Exact cutouts for 30°, 45°, and 60°
- Center Bushing: 4.00 mm solid brass precision rivet

DESK DISPLAY STAND LID & CADDY BAY

Lid features desktop easel with alignment cursor.
Stand volvelle upright while solving problems!

UNIT CIRCLE: 30° –> (cos = √3/2 ≈ 0.866, sin = 1/2)



ANGLE θ = 30° (π/6)

X = cos(30°) = √3/2 ≈ 0.866
Y = sin(30°) = 1/2 = 0.500

Pythagorean Identity:
cos²(θ) + sin²(θ) = 1
0.75 + 0.25 = 1.000

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Base Unit Circle Disc	1x	84.0 mm ⌀ (Deg & Rad)	High-Contrast White
Rotating Radian Overlay	1x	Transparent Radial Cursor	Laser-Etched PETG
Projection Cursor Arm	1x	X/Y Coordinate Windows	Clear Polycarbonate
Center Brass Pivot	1x	Low-Friction Precision Bushing	Solid Brass
Stand & Caddy Lid	1x	Angled Desk Display Stand	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 4 Volvelle Components + Storage Bin + Stand Lid = 6 Parts

QUADRANT SIGN CONVENTION

- Quadrant I (+, +): All functions positive (0° to 90°).
- Quadrant II (−, +): Sine positive, Cosine negative (90°–180°).
- Quadrant III (−, −): Tangent positive, Sine/Cos negative.
- Quadrant IV (+, −): Cosine positive, Sine negative.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Align the transparent cursor to 0° / 360°, slip the volvelle into its protective velvet caddy sleeve, and lay it flat into the box recess. Snap the stand lid shut. Lid latches firmly for silent, rattle-free carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

PYTHAGOREAN TRIGONOMETRIC IDENTITY

For any real angle θ on the unit circle: cos²(θ) + sin²(θ) = 1 identically.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 08 OF 10 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS • DUAL ROTATING CURSOR • (X, Y) PROJECTION WINDOWS

LESSON GUIDE: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

Step-by-Step Hands-On Tutorial: Angle 30° –> (cos 30° = √3/2, sin 30° = 1/2)

Grades 9–12 Trigonometry & Physics • Hands-On Classroom & Home STEM Studio Series

CARD 08 / 10 • VOLV-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/unit-circle-volvelle/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.HSF.TF.A.2 (Unit Circle)
- CCSS.MATH.CONTENT.HSF.TF.A.3 (Special Triangles)
- TEKS 111.42.c.2.P (Trig Functions)

Grades 9–12 Trigonometry & Physics

MATHEMATICAL INQUIRY

When the cursor rotates past 90° into Quadrant II, why does the horizontal X coordinate become negative while Y stays positive?

1 STEP 1: ROTATE CURSOR TO 30° (π/6 RAD)

Turn the transparent cursor arm to the 30° mark.
Look at the inner radian scale: 30° aligns exactly with π/6 rad.
Notice the right triangle projected onto the unit circle base!

2 STEP 2: READ HORIZONTAL COSINE PROJECTION

Look down at the horizontal X-axis window:
X = cos(30°) = √3 / 2 ≈ 0.8660.
This is the horizontal base length of the 30°–60°–90° triangle.

3–4 STEPS 3 & 4: READ SINE & VERIFY ON STACKCALC

3. READ VERTICAL SINE: Look at vertical Y-axis window: Y = sin(30°) = 1/2 = 0.5000.
4. VERIFY PYTHAGOREAN IDENTITY: cos²(30°) + sin²(30°) = 0.75 + 0.25 = 1.0000!
Calculator Verification: Press Unit Circle Radian Icon ([COS], [SIN] & [r][▸y,x]) and run 45 [COS] [↰] [x²] 45 [SIN] [↰] [x²] [+] → 1.

5 CORE PRINCIPLE: SINE AND COSINE ARE CIRCLE COORDINATES

Textbook right triangles only work for angles < 90°. The volvelle proves that sine and cosine are the literal (X, Y) coordinates of a point moving around a circle! In Quadrant II, cosine naturally becomes negative because X is negative, while sine remains positive!

★ EXTENSION INVESTIGATION: SPECIAL ANGLE: 60° (π/3 RAD)

Rotate cursor to 60°. Notice X and Y coordinates swap: X = cos(60°) = 1/2 = 0.500, Y = sin(60°) = √3/2 ≈ 0.866! (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem.
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CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

stackcalc.io/parts

Classroom Lab Crates: stackcalc.io/schools